

# Municipal Bond Pricing Project Final Report

Team #153: 2Legit2Quit - Jacob Burbank-Goldrich, Janarthanan Gopi, Issac J Kuo, Saswata Misra,  
Constance Julia Nething, and Angela Tran

## Introduction

Municipal (muni) bonds are debt securities issued by a state, municipality or county to finance its capital expenditures, and benefit both the issuer (school districts, universities, hospitals, etc.) in the form of lower funding costs and the investor (primarily retail) in the form of tax-free income and other tax optionality. The tax implications of municipal bonds are just one of many factors that make this type of investment complex and their pricing inefficient.

## Problem Definition

Pricing of municipal bonds is both opaque and inefficient, owing to high levels of complexity and the differing information available to interested parties. This presents the opportunity for investors to buy low and sell high.

## Objective

To provide muni bond traders with a tool to identify outliers in their existing pricing scheme which will allow them to price their bonds appropriately and gain efficiency in trading.

## Survey

The complexity in the muni bond market served as a launching point to explore how pricing is currently accounted for today. [1] [2] [3]

Previous studies have attempted to quantify the contributing factors to pricing, including default risk, credit enhancements, policy changes, city structure, and the municipality's "social capital". [4] [7] [14] [16] Studies can even contradict results of those prior when liquidity of the bond market is considered, again highlighting the interdependence of many factors. Issuer's previous decisions affect their future decisions in choosing a sale method, meaning that pricing can change over time per issuer's past choices. [11] [13] Delving into the many attributes that factor into bond pricing revealed that in addition to complexity, information asymmetry (unequal information between parties) contributes to pricing inefficiency. This is true in both the primary Initial Public Offering (IPO) market where asymmetry is the lowest and even more so in the secondary market, where asymmetry grows. We propose to use a regression to predict reasonable prices and identify outliers in order to help traders optimally price their inventory and gain insight into changing market conditions. [4] [5] [6] [7] [8] [9] [10] [11] [12] [13] [14] [15] [16]

As a result of the persistent information asymmetry and the difficulty in the current pricing process, muni bond pricing is done manually by a handful of experts. Some research has been done to investigate data driven approaches such as ridge regression, neural networks, and support vector regression. This validates the approach of using machine learning techniques to help address bond pricing; however, these approaches have not broadly taken hold. We propose to leverage visualization techniques using the results of the regression model to create a tool that allows for meaningful insights to be obtained via customizable comparisons and filtrations, which allows for complexity to be incorporated dynamically and on a case-by-case basis, reducing inefficiency and information asymmetry. [17] [18]

## Proposed Method

Previous solutions have not incorporated a configurable visual element to aide in further refining their analysis. We are proposing a solution that will combine a configurable visual component with

traditional machine learning models. This new approach will be successful because it will allow traders to visualize the pricing of their bonds against comparable bonds, which also reduces required research time. As a result, this will also increase transparency and efficiency in the market.

## Innovations

- The use of a machine learning algorithms to first create a pricing baseline, and then leveraging that baseline to identify outliers.
- The display of similar bonds in a configurable visualization with associated metrics.

## 1. Data Collection & Cleaning

To begin, 3 populations of United States (US) muni bonds as of September 24, 2020 were acquired totaling 272,283 records.

- Historical muni bond trades over the past 60 days were downloaded from the Municipal Securities Rule Making Board (MSRB).
- Competing bonds available for sale were downloaded from Bloomberg Runz.
- Current inventory of bonds was provided by team member's firm.

The bond data was then enriched with quantitative (12 attributes) and categorical (5 attributes) security level information downloaded from Bloomberg.

Using Python, descriptive statistics were then applied on the dataset and then reviewed. During the review, it was discovered that the range for adj\_spread did not make sense (spreads < -700%, mathematically infeasible) and bonds that weren't applicable to our target universe (spreads > 30%, distressed debt not traded by most traders). We decided to exclude this portion of the data (the top/bottom 1% of data points sorted by adj\_spread), and began further analysis on the remaining 266,837 records.

## 2. Data Analysis & Modeling

Our goal is to discover pricing outliers (adj\_spread) from the current inventory population. Leveraging the other 2 populations, we trained a regression model to predict spreads and then use this model to predict spreads for the inventory population and use the difference in the model predicted spread versus the actual spread to identify outliers. Using R and Python, the team evaluated different clustering and regression algorithms, which we discuss below in further detail.

## Experiments & Evaluation

### Model Evaluation

We began our experiments with the thought that we would cluster like bonds based on their attributes and then identify spread/price outliers based on the aggregate cluster pricing.

To verify the quality of clustering algorithms, we used the "eval\_spread". The eval\_spread is a high quality outside source (included in our data set) which produces accurate pricing information for all muni bonds. Bonds can and do trade very differently to the quoted eval\_spread, so that alone cannot be used for creating clusters of similar bonds. However, in an ideal cluster of similar bonds, the eval\_spread should be very close in value. More specifically for each cluster we calculated the standard deviation of the eval\_spread of each bond vs the median eval\_spread. We then calculated the weighted average standard deviation across all clusters ("C-Score").

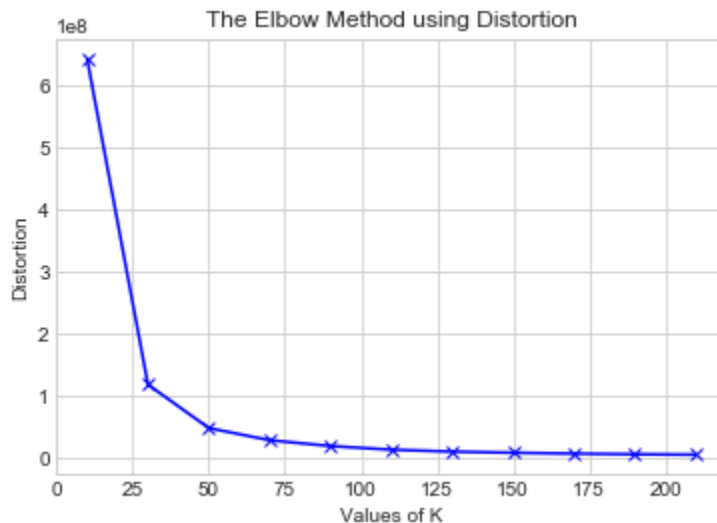
The following clustering algorithms were tested:

### Gower Distance & k-medoids

1. We planned to create a dissimilarity matrix using Gower distance for multi-scaled data.
2. Look for optimal number of clusters
3. Perform k-medoids clustering
4. Review results for functional reasonableness
5. However, there was not enough processing power to create the matrix due to the number of records and attributes. Since it could not be completed, this method was not used.

### K-Means

1. We first ran K-means with all 23 attributes (17 continuous and 6 categorical) with a  $k$ -value of 30 to form a baseline. The C-Score was 0.3402.
2. We then created an elbow plot of distortions vs  $k$ -values to find the optimal  $k$  (50) and reran the analysis producing a C-Score of 0.3074.



3. Using the knowledge that k-means does not handle categorical variables well, we replaced two of the categorical variables with continuous values. State is a meaningful categorical variable because municipal bonds are free from both federal and state income taxes. Therefore, the difference in tax regimes by state as well as median household incomes ("MHI) impacts the demand for a particular bond. For example, WA has a high MHI but 0% income tax, so WA domiciled bonds carry no tax benefit to the WA resident and reduce demand for those bonds. AR has a relatively high tax rate at 6.6%, but its wealth levels are low which causes a low demand for tax free bonds. CA on the other hand has both a high tax rate (13.3%) and also a high MHI which increases demand and therefore causes its bonds to trade at lower spreads/higher prices. In the same way Sector is meaningful because of the historical default risk of each sector. We substituted the historical default rates for each sector and reran the analysis producing a C-score of 0.3048.
4. We then examined whether the results could be improved by reducing the number of variables to only those that domain knowledge suggested were the most important. Using only six variables produced a worse C-score of 0.3352.
5. We hypothesized that k-means could benefit from a pre-assignment of the relative importance of the variables. We used domain knowledge to construct a weight vector and took multiplied the standardized and produced a C-score of 0.2874. We concluded that even if we could get the 0.0484 benefit with the weight vector on the subset of variables in step 3 that the clustering would not be accurate enough for our purposes.

### PCA + DBSCAN:

1. Run PCA on the quantitative data
2. Plot the top 2 principal components
3. Run DBSCAN on the scatterplot to identify clusters
4. Multiple runs on each state aggregate bond data using varying epsilon and n values until one main group and noise group were outputted by DBSCAN.
5. The results of DBSCAN demonstrated that outliers could be detected from entire bond population but not on similar bonds.
6. We concluded that the results were not satisfactory, and it became apparent that some of the categorical data was necessary in order to properly cluster similar bonds.

**Kamila:**

1. The Kamila package in R was experimented due to its ability to better handle categorical variables. Unfortunately, after several iterations of different combinations of attributes, the results were very similar. The model would create multiple small clusters consisting of 1 to 10 bonds and a few very large clusters consisting over 100,000 bonds. These results did not meet the business purpose when visualized and therefore, not used.

**FAMD Clustering:**

After researching other methods and algorithms to cluster mixed data, FAMD (Factor Analysis of Mixed Data) appeared to be a prime candidate for the task. FAMD is a combination of PCA and Multiple Correspondence Analysis (MCA) for categorical data.

1. Trial different selections of core bond attributes to cluster by, selecting only from historical + competing bonds
2. Interpret the clusters based on metrics like eval\_spread.
3. Initial runs appeared promising for clustering neatly. Due to time constraints and the success of the regression model, a full analysis of clusters was not completed. However, we plan to use this method for v2.0 of the tool.

**Regression: Chosen as the final method**

1. We first trained a multivariate linear regression model with spread as the result. We achieved a 0.0522 Mean Squared Error ("MSE"). A Box Cox transformation improved the MSE to 0.0513.
2. We tested every combination of pairs of the predictors and kept the ten that produced the highest  $R^2$ . This improved the MSE to 0.0389.
3. We tested polynomial transformations up to degree  $n=5$  again keeping the 10 that produced the highest  $R^2$ . This improved the MSE to 0.0322.
4. We used Ridge regression to regularize the model, but the MSE worsened to 0.355 so we did not include this in the final model.
5. We used Lasso for variable selection, but the MSE worsened to 0.378 so we did not include this in the final model.
6. We fit a Random Forest regression model which improved the MSE to 0.0246.
7. We clustered the bonds using K-means and then trained individual regression models on each of the clusters. The MSE underperformed the Random Forest at 0.0332.
8. We fit an Adaboost Decision Tree Regressor and the MSE improved to 0.0239 with a Root MSE at 0.1546. This result compared favorably to the best C-score from the clustering algorithms at 0.2468. As a result, Adaboost Decision Tree Regressor was chosen as the final model.
9. We trained on 80% of the Historical and Competing populations and predicted spread on the remaining 20% set aside for testing. We calculated the standard deviation (std\_dev) of the difference in model predicted spread versus the actual spread to obtain a baseline level of noise one could expect in this population and model. We then used the model to predict spreads on the Inventory population and computed a relative "sigma" (actual spread - model

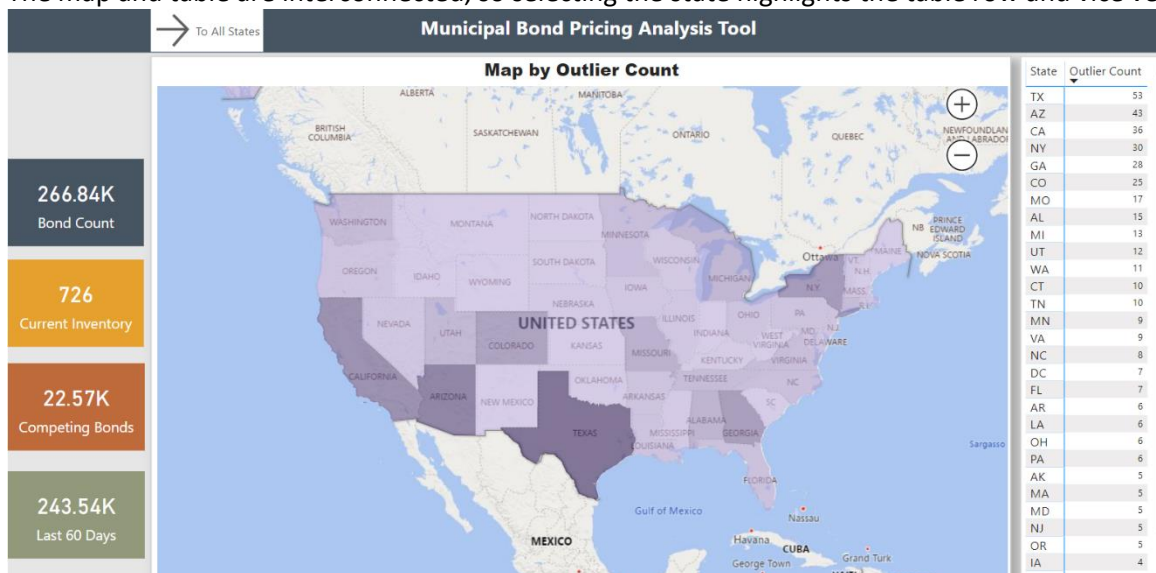
spread)/std\_dev. This is a measure of the deviance of the actual spread vs the model spread of each bond relative to the noise in the trained model.

### 3. Visualization & User Interface (UI)

We identified a suitable regression algorithm and used it to compute an outlier score which is a measure of the deviation in the model predicted versus the spread of the inventory against the historical and competing populations.

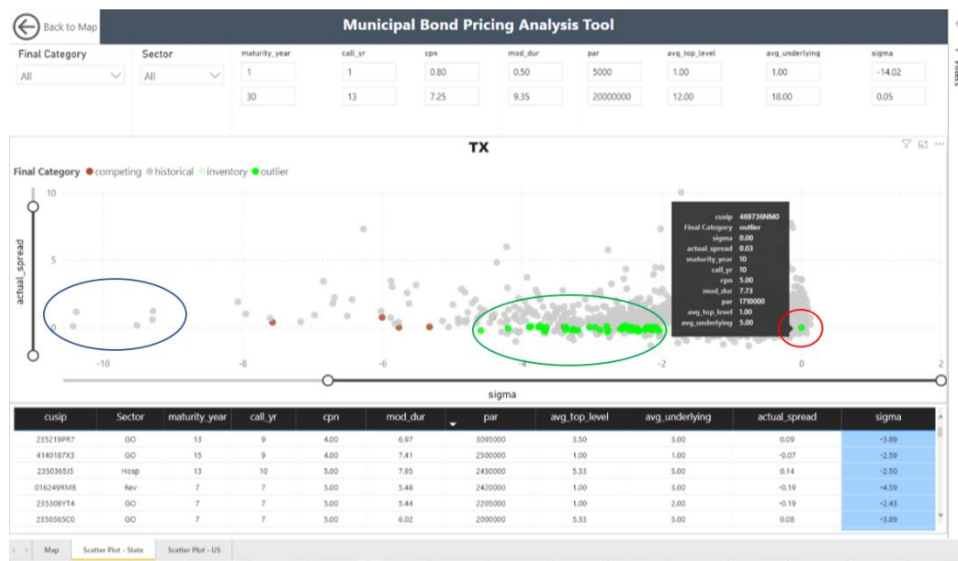
#### Initial Visualization

The UI begins with a choropleth map of the U.S. color coded by the count of outliers in each state. To assist the user (trader) in quickly identifying the most important cluster to review for pricing outliers, a table of states and outlier scores sorted in descending order are provided on the right side of the page. The map and table are interconnected, so selecting the state highlights the table row and vice versa.



#### Interactivity

Upon right-clicking on a state from the map or table, the user can drill through to a scatterplot of bonds with the outlier score on the x-axis and spreads on the y-axis. A table listing the bonds ranked by highest outlier score is displayed to the right. Clicking or hovering over a bond in the list displays statistics about that bond. Additional filters are also available for the user to narrow down the number of bonds displayed on the scatterplot.



The power of the visualization is to enable the user to sift through over 260,000 bonds and quickly identify those from their current inventory that need adjusting:

- Bonds that are priced too high/spread too low (negative sigma) and may never sell at that spread/price (green circle)
- Bonds that are priced too low/spread too high (positive sigma) and gains are not maximized (red circle)

An added benefit that was not anticipated is that visualizing outliers in the Historical population is useful as well. If a number of Historical trades were done at outlier type prices/spreads and the user can identify similar attributes among them, this might present a future investment opportunity (blue circle).

## Conclusions

### Impact & Success Criteria

The success of this tool is measured by visually demonstrating various opportunities for which the price of a bond within the inventory population should be priced lower in order to increase the chances of sale or higher to increase the profitability, reducing the likelihood that the price a trader sells at would be an outlier in that group. We were able to display over 260,000 bonds in a way that shows their relative pricing, but also highlight the handful that the user may want to investigate further. Overall, we were pleased with the results of our analytics and the actionable data points that the visualization produces.

### Distribution of Team Member Effort

|                         |     |
|-------------------------|-----|
| Jacob Burbank-Goldrich  | 21% |
| Issac J Kuo             | 21% |
| Constance Julia Nothing | 21% |
| Angela Tran             | 21% |
| Saswata Misra           | 10% |
| Janarthanan Gopi        | 6%  |

## References

- [1] H. Galper, K. Rueben, R. Auxier and A. Eng, "MUNICIPAL DEBT: WHAT DOES IT BUY AND WHO BENEFITS?," *National Tax Journal*, no. 67.4, pp. 901-24, 2014.
- [2] D. B. Vukovic, M. Maiti, D. Kochetkov and A. Bystryakov, "How Attractive Are Municipal Bonds for the Passive Competitiveness: The Case of Immunization of Municipal Bonds," *Competitiveness Review*, Vols. ahead-of-print, 2020.
- [3] A. Kalotay and C. D. Howard, "The Tax Option in Municipal Bonds," *Journal of Portfolio Management*, no. 4.02, pp. 94-102, 2014.
- [4] M. Schwert, "Municipal Bond Liquidity and Default Risk," *The Journal of Finance (New York)*, no. 72.4, pp. 1683-722, 2017.
- [5] J. Peng and P. F. Brucato, "An Empirical Analysis of Market and Institutional Mechanisms for Alleviating Information Asymmetry in the Municipal Bond Market," *Journal of Economics and Finance*, no. 28.2, pp. 226-38, 2004.
- [6] M. Ivonchyk, "The Impact of Dodd-Frank on True Interest Cost of Municipal Bonds: Evidence From California," *Public Budgeting & Finance*, no. 39.1, pp. 3-23, 2019.
- [7] J. Dove, "Local Government Type and Municipal Bond Ratings: What's the Relationship?," *Applied Economics*, no. 49.24, pp. 2339-351, 2017.
- [8] D. V. Denison, W. Yan and Z. Zhao, "Is Management Performance a Factor in Municipal Bond Credit Ratings?," *Public Budgeting & Finance*, no. 27.4, pp. 86-98, 2007.
- [9] A. Ang, R. C. Green, F. A. Longstaff and Y. Xing, "Advance Refundings of Municipal Bonds," *The Journal of Finance (New York)*, no. 72.4, pp. 1645-682, 2017.
- [10] C. Partridge and F. R. Medda, "The Evolution of Pricing Performance of Green Municipal Bonds," *Journal of Sustainable Finance & Investment*, no. 10.1, pp. 44-64, 2020.
- [11] G. Liu, "Self-selection Bias or Decision Inertia? Explaining the Municipal Bond "competitive Sale Dilemma"," *Journal of Public Budgeting, Accounting & Financial Management*, no. 30.1, pp. 86-106, 2018.
- [12] C. Downing and F. Zhang, "Trading Activity and Price Volatility in the Municipal Bond Market," *The Journal of Finance (New York)*, no. 59.2, pp. 899-931, 2004.
- [13] G. Liu, "The Effect of Sale Methods on the Interest Rate of Municipal Bonds: A Heterogeneous Endogenous Treatment Estimation," *Public Budgeting & Finance*, no. 38.2, pp. 81-110, 2018.
- [14] C. Rosa, "Municipal Bonds and Monetary Policy: Evidence from the Fed Funds Futures Market," *The Journal of Futures Markets*, no. 34.5, pp. 434-50, 2014.

- [15] A. L. Chun, E. Namvar, X. Ye and F. Yu, "Modeling Municipal Yields With (and Without) Bond Insurance," *Management Science*, no. 65.8, pp. 3694-713, 2019.
- [16] P. Li, L. Tang and B. Jaggi, "Social Capital and the Municipal Bond Market," *Journal of business ethics*, vol. 153, pp. 479-501, 2016.
- [17] G. H. Dash, N. Kajiji and D. Vonella, "The Role of Supervised Learning in the Decision Process to Fair Trade US Municipal Debt," *EURO Journal on Decision Processes*, no. 6.1, pp. 139-68, 2018.
- [18] N. Raman and J. Leidner, "Municipal Bond Pricing: A Data Driven Method," *International Journal of Financial Studies*, no. 6.3, 2018.